Deep Learning for Top-Quark Jet Tagging

A Data-Efficiency and Probability-Calibration Study on the 14 TeV Reference Dataset

A Final-Year Project Report Submitted in Partial Fulfilment of the Requirements for the Degree of

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Submitted by

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DECLARATION BY THE CANDIDATE

I, **Bibi Naifa, Mah Lacka, Kanwal Gul & Firdos (BS Physics, Final-Year Project (2022–2026))**, hereby declare that the work presented in this thesis entitled **“Deep Learning for Top-Quark Jet Tagging: A Data-Efficiency and Probability-Calibration Study on the 14 TeV Reference Dataset”** is the result of the work performed by me in the **Department of Natural Sciences, Begum Nusrat Bhutto Women University, Sukkur**, under the supervision of **Dr. Ram Chand** and that it has not been submitted elsewhere for any degree or diploma. I further declare that:

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3. All ethical guidelines prescribed by the university for research have been followed in this study.

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CERTIFICATE

This is to certify that the work presented in this thesis, entitled **"Deep Learning for Top-Quark Jet Tagging: A Data-Efficiency and Probability-Calibration Study on the 14 TeV Reference Dataset"**, was carried out by **Bibi Naifa**, **Mah Lacka**, **Kanwal Gul** and **Firdos** as a group Final-Year Project towards the degree of **Bachelor of Science in Physics**, under my supervision at the Department of Natural Sciences, The Begum Nusrat Bhutto Women University, Sukkur. To the best of my knowledge, this is an original piece of work and no part of it has been submitted for any other degree or qualification.

Dr. Ram Chand
Thesis Supervisor

May 28, 2026

Prof. Dr. Ram Chand
Chairperson, Dept. of Natural Sciences

Prof. Dr. Ram Chand
Dean of Faculty

To our families, whose patience and encouragement sustained us, and to every student who believes that world-class physics can be done with curiosity, open data, and a modest laptop.

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We owe our deepest gratitude to our supervisor, **Dr. Ram Chand**, who proposed this project, taught us the physics of jets and the craft of machine learning, and patiently guided every stage of the work. His insistence on reproducibility and honest reporting shaped this thesis throughout.

We thank the **Department of Natural Sciences** at The Begum Nusrat Bhutto Women University, Sukkur, for the academic environment and computing support that made this study possible, and our teachers and classmates for their encouragement and feedback.

This work would not have been possible without the open-science ecosystem of particle physics and machine learning. We gratefully acknowledge G. Kasieczka, T. Plehn, J. Thompson and M. Russel for releasing the Top-Quark Tagging Reference Dataset on Zenodo, and the developers of NumPy, PyTorch, XGBoost, scikit-learn, Matplotlib and Google Colaboratory, whose free tools carried out the heavy computation in this project.

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Abstract

Identifying the hadronic decays of highly boosted top quarks — "top tagging" — is a benchmark problem in collider physics and a proving ground for modern machine learning. This thesis compares four classifiers on the public Top-Quark Tagging Reference Dataset of 14 TeV proton–proton collisions: a gradient-boosted decision tree (BDT) on engineered jet-substructure variables, a convolutional neural network (CNN) on jet images, the ParticleNet graph network, and the Particle Transformer. Beyond reproducing the usual ranking, the study asks two questions of direct practical value: how much training data each architecture needs (data efficiency), and whether its output scores can be trusted as probabilities (calibration). Using a reproducible pipeline trained on a free cloud GPU, we sweep the training set from 5×10^3 to 5×10^4 jets and measure test AUC, background rejection at fixed signal efficiency, the Expected Calibration Error (ECE), and the effect of temperature scaling. The Particle Transformer attains the best discrimination (AUC $= 0.978$, background rejection ≈ 136 at 50% signal efficiency), followed by ParticleNet, the CNN and the BDT, reproducing the published ordering. The data-efficiency study reveals a clear crossover: the BDT is strongest at the smallest training size but is overtaken by the deep models as data grows, the Particle Transformer rising from weakest to strongest. The calibration study shows that ranking and calibration are decoupled — the jet-image CNN is the worst-calibrated model (ECE $= 0.078$) despite its mid-pack ranking, whereas the top-ranked transformer is comparatively well-calibrated; a single-parameter temperature rescaling corrects all models except the CNN. The complete pipeline, trained models and documents are released as open source.

Keywords: top tagging, jet substructure, deep learning, ParticleNet, Particle Transformer, data efficiency, probability calibration.

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1 Introduction

1.1 | Background

The Large Hadron Collider (LHC) at CERN collides protons at a centre-of-mass energy of up to 14 TeV and records the debris with detectors such as ATLAS and CMS. Most of the interesting particles produced in these collisions are not observed directly; instead they decay into sprays of hadrons that are reconstructed into collimated bundles called *jets* (Cacciari et al., 2008; Salam, 2010). Reading the internal structure of a jet — its *substructure* — is the key to deciding which kind of particle produced it (Kogler et al., 2019; Larkoski et al., 2020).

The top quark is the heaviest known elementary particle, with a mass of $m_t \simeq 172.76\,\mathrm{GeV}$ (Workman and others , Particle Data Group). It decays almost exclusively through $t \to Wb$, and when the W boson decays hadronically the full chain becomes $t \to Wb \to q\bar{q}'b$. If the top quark is produced with a large transverse momentum (it is then said to be *boosted*), the three resulting quarks are captured inside a single large-radius jet that carries a characteristic three-pronged energy pattern and a mass near m_t. Distinguishing such "top jets" from the vastly more common jets initiated by light quarks and gluons (Quantum Chromodynamics, QCD, jets) is known as *top tagging*, and it is one of the canonical classification problems of collider physics (Kasieczka et al., 2019b).

Over the last decade, top tagging has become a benchmark for machine learning in physics. Early approaches engineered physically motivated variables such as the jet mass, N-subjettiness (Thaler and Van Tilburg, 2011) and energy-correlation functions (Larkoski et al., 2013) and fed them to boosted decision trees or shallow networks. The field then moved to deep models that act directly on the low-level constituents: convolutional networks on pixelated "jet images" (de Oliveira et al., 2016; Komiske et al., 2017), graph networks such as ParticleNet (Qu and Gouskos, 2020), and most recently attention-based models such as the Particle Transformer (Qu et al., 2022). On the standard reference dataset (Kasieczka et al., 2019a), these models reach areas under the ROC curve (AUC) above 0.98, and a community comparison (Kasieczka et al., 2019b) established the relative ordering that is now treated as the state of the art.

1.2 | Problem Statement

The published top-tagging literature is almost entirely concerned with the *best achievable* discrimination, quoted at the full training-set size of 1.2×10^6 jets and on large GPU clusters. Two questions of direct practical importance are rarely addressed together:

1. **Data efficiency.** Generating labelled Monte Carlo training samples is computationally expensive. How much does each architecture's performance depend on the amount of training data, and does the ranking of the models change in the small-data regime that a student or a small group actually works in?

2. **Probability calibration.** A high AUC only certifies that a model *ranks* signal above background. Many downstream physics analyses instead use the classifier score as a *probability* (for example, in a likelihood fit). Are the scores produced by these models trustworthy probabilities, and if not, can they be cheaply corrected?

This thesis addresses both questions on the public reference dataset using a single, reproducible pipeline that can be run end-to-end on free hardware.

1.3 | Aim and Research Objectives

The overall **aim** of this project is to build a transparent, reproducible top-tagging pipeline and use it to study the data efficiency and probability calibration of four representative classifiers, rather than to chase a new record AUC.

The specific **objectives** are:

1. **To implement** a unified pipeline that trains and evaluates four architectures — a gradient-boosted decision tree (BDT) on engineered features, a convolutional neural network (CNN) on jet images, the ParticleNet graph network, and the Particle Transformer — under identical preprocessing, data splits and random seeds.

2. **To reproduce** the established performance ordering on the Top-Quark Tagging Reference Dataset and report AUC, accuracy and background rejection at fixed signal efficiency.

3. **To quantify data efficiency** by re-training each model from scratch at four training-set sizes spanning a decade (5×10^3 to 5×10^4 jets) and measuring how performance scales.

4. **To assess probability calibration** using reliability diagrams and the Expected Calibration Error (ECE), and to test whether a single-parameter temperature rescaling restores calibration.

5. **To make the work fully reproducible**, releasing the code, trained models and documents as open source and providing a cloud-GPU workflow that any student can follow.

1.4 | Research Questions

1. Does the literature ordering (Particle Transformer > ParticleNet > CNN > BDT) hold when all four models are trained identically on a modest data and compute budget?

2. How does each model's discrimination scale with the training-set size, and is there a regime in which the simple BDT is preferable to the deep models?

3. Are the classifier scores well-calibrated probabilities, and does temperature scaling fix any miscalibration?

1.5 | Significance of the Study

This study is significant for three reasons. First, it provides a clear, honest, reproducible reference implementation of four top-tagging architectures that can be used for teaching and as a baseline for future projects at the host institution and beyond. Second, the data-efficiency results give practitioners concrete guidance on which model to choose under a limited Monte Carlo budget — a situation common outside the large LHC collaborations. Third, the calibration analysis highlights a subtle but important point often overlooked in physics machine-learning papers: a model with the best ranking power is not necessarily the one whose scores can be read as probabilities. Demonstrating that the entire study can be carried out on a free cloud GPU also lowers the barrier to entry for students in resource-constrained settings.

1.6 | Scope and Limitations

The study is deliberately bounded. We use stratified subsamples of up to 5×10^4 jets per split rather than the full 1.2×10^6-jet dataset, so the absolute AUC values are a little below the published saturation values; the *trends* and *rankings*, not the absolute records, are the object of study. The dataset itself is simulated with a single detector parametrisation and contains no pile-up, so questions of detector robustness and high-luminosity performance are out of scope. The four architectures are representative rather than exhaustive, and each is trained with a fixed, modest hyper-parameter configuration rather than an exhaustive search. These limitations are revisited in Chapter 5.

1.7 | Thesis Organization

The remainder of this thesis is organised as follows. Chapter 2 reviews the physics of jets and top quarks and the machine-learning approaches to tagging. Chapter 3 describes the dataset, the feature engineering, the four model architectures and the experimental protocol. Chapter 4 presents the headline benchmark, the data-efficiency sweep and the calibration analysis, and discusses their meaning. Chapter 5 summarises the findings, states the limitations and proposes future work. Appendix A gives reproduction instructions and selected implementation details.

2

Literature Review

This chapter sets the physical and computational context for the project. It reviews the Standard Model and the role of the top quark, the formation and substructure of jets, the engineered observables used in classical tagging, and the deep-learning architectures that define the current state of the art.

2.1 | The Standard Model and the Top Quark

The Standard Model (SM) of particle physics describes the electromagnetic, weak and strong interactions of quarks and leptons through gauge symmetry $SU(3)_C \times SU(2)_L \times U(1)_Y$ (Workman and others , Particle Data Group). The top quark, the up-type quark of the third generation, is the heaviest known elementary particle, $m_t \simeq 172.76\,\mathrm{GeV}$. Because of this large mass its lifetime is shorter than the timescale of hadronisation, so it decays as a bare quark, almost always via $t \to Wb$. The W boson subsequently decays either leptonically ($W \to \ell\nu$) or hadronically ($W \to q\bar{q}'$). The hadronic channel, $t \to Wb \to q\bar{q}'b$, is the one relevant to this thesis: it produces three quarks that, for a boosted top, merge into one large-radius jet.

2.2 | Jets and Jet Reconstruction

Coloured partons cannot exist freely; they fragment and hadronise into collimated streams of colourless hadrons. These streams are reconstructed into jets by sequential-recombination algorithms, most commonly the anti-k_t algorithm (Cacciari et al., 2008) as implemented in FastJet (Cacciari et al., 2012). For boosted-object studies a large radius parameter ($R = 0.8$ or 1.0) is used so that the entire decay is contained in a single jet. A boosted top jet then differs from an ordinary QCD jet in three observable ways: it is more massive (its mass peaks near m_t), it has a three-pronged rather than one-pronged energy flow, and it contains more constituents. Exploiting these differences is the essence of jet-substructure analysis (Kogler et al., 2019; Larkoski et al., 2020).

2.3 | Engineered Substructure Observables

Classical taggers compress the jet into a handful of physically motivated numbers. The most important are:

- **Jet mass** — the invariant mass of the summed constituent four-momenta, which discriminates the heavy top from light QCD jets.
- **N-subjettiness** τ_N and its ratios $\tau_{21} = \tau_2/\tau_1$, $\tau_{32} = \tau_3/\tau_2$, which quantify how well the radiation is described by N subjet axes (Thaler and Van Tilburg, 2011). A genuine three-prong top jet has small τ_{32}.
- **Energy-correlation functions** and the ratios C_2, D_2, which capture n-point energy correlations without explicit subjet finding (Larkoski et al., 2013, 2014).

Fed to a boosted decision tree (BDT), typically implemented with XGBoost (Chen and Guestrin, 2016), these observables already provide strong discrimination and remain a competitive, interpretable baseline.

2.4 | Deep Learning on Jets

The deep-learning era of tagging replaced hand-engineered features with representations learned directly from the constituents. Three families are relevant here.

2.4.1 | Jet Images and Convolutional Networks

Pixelating the constituents in the rapidity–azimuth plane turns a jet into an image that a convolutional neural network (CNN) can classify, in close analogy with calorimeter readout (de Oliveira et al., 2016; Komiske et al., 2017). Image methods were the first deep approaches to surpass engineered-feature BDTs, but the binning discards angular resolution and breaks the natural permutation symmetry of the constituents.

2.4.2 | Particle Clouds and Graph Networks

Treating a jet as an unordered set — a "particle cloud" — preserves the full constituent information. ParticleNet (Qu and Gouskos, 2020) adapts the Dynamic Graph CNN (Wang et al., 2019) to jets, building a k-nearest-neighbour graph in feature space and applying EdgeConv operations that are invariant to constituent ordering. This and related set/graph approaches (Komiske et al., 2019; Moreno et al., 2020) substantially improved tagging performance.

2.4.3 | Attention and the Particle Transformer

The Transformer architecture (Vaswani et al., 2017) models pairwise interactions through self-attention. The Particle Transformer (Qu et al., 2022) augments standard attention with physically motivated pairwise features (angular distance, k_t, momentum fraction and invariant mass) injected as an attention bias, and currently holds the top of the reference-dataset leaderboard (Kasieczka et al., 2019b). Related physics-informed and Lorentz-equivariant designs continue to push the frontier (Bogatskiy et al., 2020; Gong et al., 2022).

2.5 | Probability Calibration in Machine Learning

A classifier is *calibrated* if, among samples it assigns score p, a fraction p are truly positive. Guo et al. (2017) showed that modern deep networks, although accurate, are frequently *over-confident* and that a simple post-hoc *temperature scaling* — dividing the logits by a single learned scalar T — often restores calibration. Calibration is measured by the Expected Calibration Error (ECE) and visualised with reliability diagrams (Niculescu-Mizil and Caruana, 2005). Although calibration is standard in the broader ML community, it is seldom reported in physics-tagging papers, which is precisely the gap this thesis explores.

2.6 | Summary and Gap

The literature establishes both a strong set of engineered observables and a clear hierarchy of deep models, all benchmarked at the full data and compute budget. What is missing, and what this thesis provides, is a controlled, reproducible comparison of *data efficiency* and *probability calibration* across these architectures in the modest-budget regime that is realistic for most students and small groups.

3

Methodology

This chapter describes how the study was carried out: the dataset, the preprocessing and feature engineering, the four model architectures, the training protocol, and the metrics used for evaluation, data efficiency and calibration. The complete implementation is released as open source.[1]

3.1 | Dataset

We use the public *Top-Quark Tagging Reference Dataset* (Kasieczka et al., 2019a), the community-standard benchmark for this problem (Kasieczka et al., 2019b). It contains 2×10^6 simulated jets from 14 TeV proton–proton collisions, generated with Pythia 8 (Sjöstrand et al., 2015) and a Delphes fast detector simulation (de Favereau et al., 2014) of an ATLAS-like detector. Signal jets come from boosted hadronic top decays; background jets come from QCD dijet production. Jets are clustered with the anti-k_t algorithm (Cacciari et al., 2008) with radius $R = 0.8$ and transverse momentum $550 < p_T < 650\,\mathrm{GeV}$, and each jet is stored as up to 200 constituent four-momenta with a binary label (top vs. QCD).

For computational tractability we draw stratified subsamples of up to 5×10^4 jets per split (training, validation, test), keeping the roughly 50/50 signal-to-background balance of the full dataset. The deep models use the leading 100 constituents per jet, ordered by p_T.

[1] `https://github.com/ramc77/fyp-jet-tagging`

3.2 | Preprocessing and Feature Engineering

3.2.1 | Per-constituent features

From each constituent four-momentum (E, p_x, p_y, p_z) we compute the transverse momentum p_T, pseudorapidity η and azimuth ϕ, and the coordinates relative to the p_T-weighted jet axis, $(\Delta\eta, \Delta\phi)$. The resulting seven-dimensional per-constituent feature vector

$$\mathbf{x} = \left(p_T,\ \eta,\ \phi,\ E,\ \Delta\eta,\ \Delta\phi,\ \log p_T\right) \tag{3.1}$$

is standardised feature-wise using statistics computed on the training split only.

3.2.2 | Engineered jet-level features (for the BDT)

For the BDT baseline we compute thirteen high-level observables: the jet mass, p_T and η; the constituent multiplicity; the jet width; the leading and sub-leading p_T fractions; the N-subjettiness variables τ_1, τ_2, τ_3 and ratios τ_{21}, τ_{32} (Thaler and Van Tilburg, 2011); and the energy-correlation ratio C_2 (Larkoski et al., 2013). Their distributions for signal and background are shown in Chapter 4 and behave exactly as physics expects, validating the pipeline.

3.2.3 | Jet images (for the CNN)

For the CNN, constituents are binned into 40×40 images in the $(\Delta\eta, \Delta\phi)$ plane with three channels: the p_T sum, the constituent multiplicity, and the p_T^2 sum per pixel. Each channel is normalised by its per-image maximum.

3.2.4 | Pairwise interaction features (for the Transformer)

For the Particle Transformer we additionally compute, on the fly, four pairwise quantities for every constituent pair (i, j): $\ln \Delta R_{ij}$, $\ln k_{T,ij}$, the momentum fraction z_{ij}, and $\Delta\eta_{ij}$, following Qu et al. (2022).

3.3 | Model Architectures

3.3.1 | Boosted decision tree (BDT)

A gradient-boosted decision tree trained with XGBoost (Chen and Guestrin, 2016) on the thirteen engineered features, with up to 500 boosting rounds, maximum depth 7, learning rate 0.1 and column/row subsampling 0.8, with early stopping on validation AUC. It is the interpretable, classical baseline.

3.3.2 | Convolutional neural network (CNN)

A compact ResNet-style network (He et al., 2016) with three residual blocks ($64 \to 128 \to 256$ channels), batch normalisation, global average pooling and a fully connected head, acting on the $40 \times 40 \times 3$ jet images.

3.3.3 | ParticleNet

A from-scratch implementation of ParticleNet (Qu and Gouskos, 2020), using three EdgeConv blocks (Wang et al., 2019) with a $k = 16$ dynamic nearest-neighbour graph in $(\Delta\eta, \Delta\phi)$ space, followed by global pooling and a fully connected classifier. It treats the jet as a permutation-invariant particle cloud.

3.3.4 | Particle Transformer

A self-attention model (Qu et al., 2022) with three encoder blocks, multi-head attention, and the four pairwise features injected as an additive attention bias. A learnable class token aggregates the sequence for classification, and its attention weights provide an interpretability handle.

3.4 | Training Protocol

All deep models are trained by minimising binary cross-entropy with the Adam/AdamW optimiser (Kingma and Ba, 2015; Loshchilov and Hutter, 2019), with learning rates of 10^{-3}–10^{-4}, a plateau learning-rate scheduler, gradient clipping and early stopping on validation AUC. Training was performed on a free Google Colaboratory T4 GPU; the same code runs on CPU. To make long runs robust, the pipeline checkpoints after every epoch and resumes automatically, so a session can be interrupted and continued without loss of progress.

3.5 | Evaluation Metrics

3.5.1 | Discrimination

We report the area under the ROC curve (AUC), the accuracy at a score threshold of 0.5, and the *background rejection* $1/\varepsilon_B$ at fixed signal efficiencies $\varepsilon_S = 0.5$ and 0.3, which is the figure of merit most relevant to physics analyses. Uncertainties on the AUC are estimated by bootstrap resampling of the test set.

3.5.2 | Data efficiency

Each model is retrained from scratch at $N_{\text{train}} \in \{5{,}000, 10{,}000, 25{,}000, 50{,}000\}$ jets, with all other settings fixed, and the resulting test AUC and background rejection are plotted against N_{train} to expose how each architecture scales.

3.5.3 | Calibration

We measure the Expected Calibration Error

$$\text{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} \left| \operatorname{acc}(B_m) - \operatorname{conf}(B_m) \right|, \tag{3.2}$$

where the predictions are partitioned into M equal-width score bins B_m and acc and conf are the bin accuracy and mean confidence (Guo et al., 2017). We complement the ECE with reliability diagrams, and we fit a single *temperature* T on the validation set by minimising the negative log-likelihood of $\sigma(z/T)$, where z are the logits, reporting the ECE before and after this rescaling.

4

Results and Discussion

This chapter presents the results in three parts that mirror the objectives of Chapter 1: a physics validation and headline benchmark (Section 4.1), the data-efficiency study (Section 4.2), and the calibration study (Section 4.3). The interpretability of the models is examined in Section 4.4 and the findings are summarised in Section 4.5.

4.1 | Physics Validation and Benchmark

4.1.1 | The physics is correct

Before comparing classifiers we confirm that the preprocessing reproduces the expected physics. Figure 4.1 shows the thirteen engineered observables for signal and background. The top-jet mass peaks near $m_t \simeq 173\,\mathrm{GeV}$ while QCD jets are light; top jets have more constituents and smaller τ_{32}, the signature of three-prong structure. Figure 4.2 shows the averaged jet images, in which the characteristic three prongs of the top decay are visible in the signal−background difference. These textbook features confirm that the pipeline is physically sound.

4.1.2 | Model comparison

Table 4.1 reports the headline performance of the four models, and Figure 4.3 shows the ROC curves and background rejection versus signal efficiency. The Par-

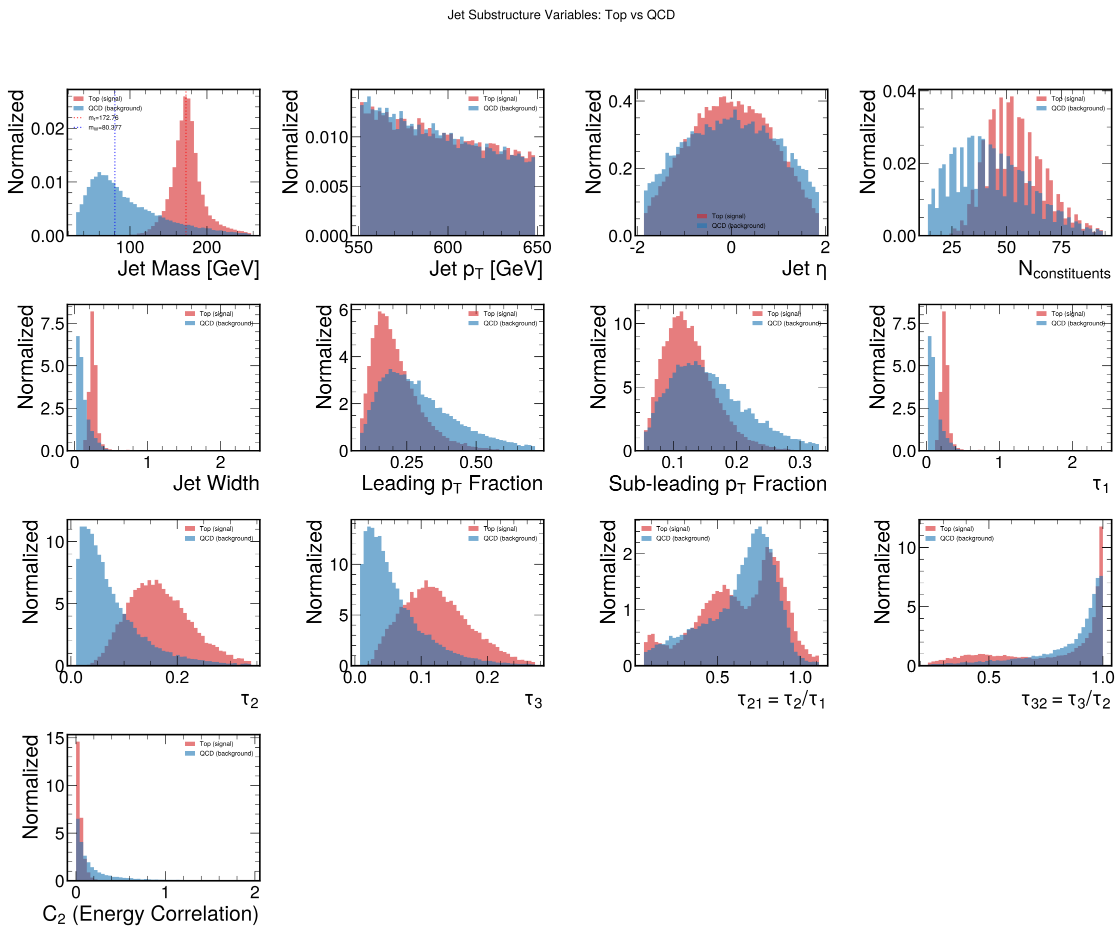

Figure 4.1: Distributions of the thirteen engineered jet-substructure observables for top (signal) and QCD (background) jets on the test set. The jet mass peaks at m_t for top jets; τ_{32} and the constituent multiplicity also separate the classes, as expected from physics.

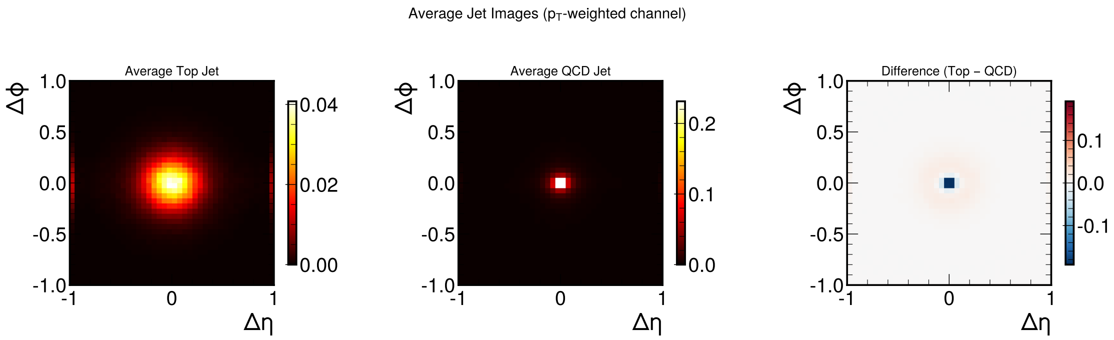

Figure 4.2: Average p_T-weighted jet images for top jets (left), QCD jets (centre) and their difference (right). The three-prong structure of the hadronic top decay is visible in the difference panel.

ticle Transformer achieves the best discrimination (AUC $= 0.978$, background rejection $1/\varepsilon_B \approx 136$ at $\varepsilon_S = 0.5$), followed by ParticleNet, the CNN and the BDT. This reproduces the ordering reported in the community comparison (Kasieczka et al., 2019b); the absolute values sit slightly below the published saturation values because we train on a subset rather than the full 1.2×10^6 jets.

Table 4.1: Comparison of jet-classification models on the Top-Quark Tagging benchmark. R_{50} and R_{30} are the background rejections $1/\varepsilon_B$ at signal efficiencies $\varepsilon_S = 0.5$ and 0.3.

Model	AUC	Acc.	R_{50}	R_{30}	Params
BDT (XGBoost)	0.9697	0.9114	93	287	164 trees
CNN (Jet Images)	0.9723	0.9155	111	413	1.26M
ParticleNet	0.9756	0.9199	130	413	0.37M
Particle Transformer	0.9775	0.9246	136	476	0.43M

Figure 4.4 shows the classifier-score distributions and Figure 4.5 the confusion matrices at a threshold of 0.5; both confirm clean signal/background separation for all models, sharpest for the deep architectures.

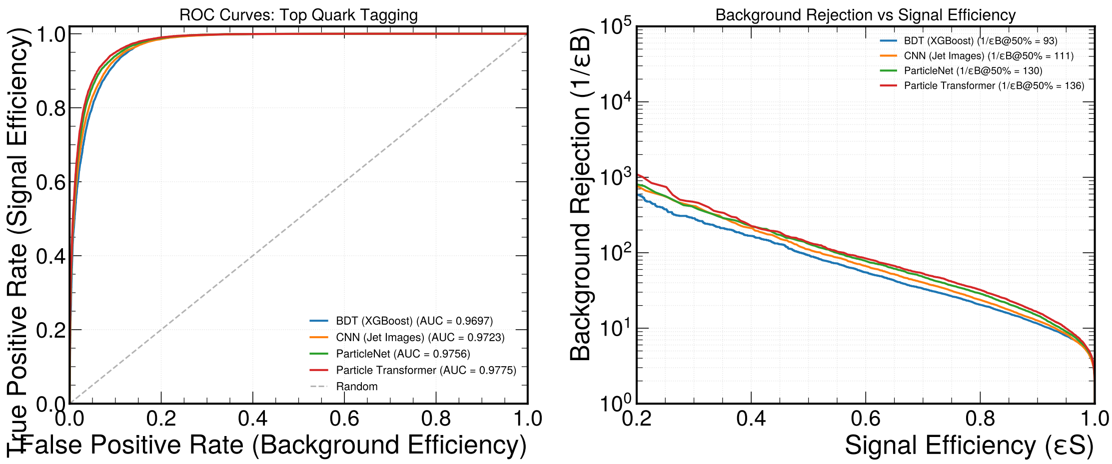

Figure 4.3: Left: ROC curves for the four models on the test set. Right: background rejection $1/\varepsilon_B$ as a function of signal efficiency ε_S. The Particle Transformer (red) is best across the range.

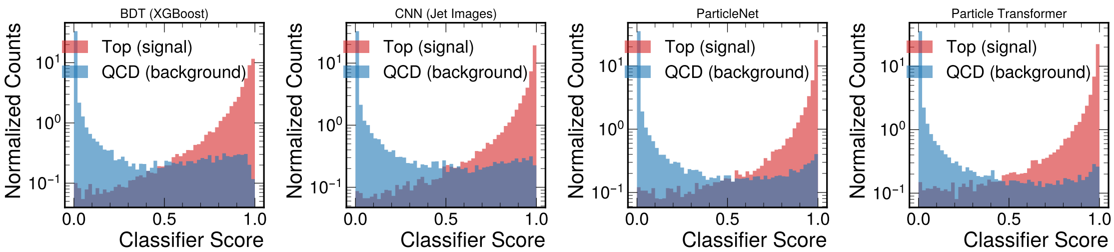

Figure 4.4: Classifier-score distributions for top (red) and QCD (blue) jets. Well-separated peaks near 0 and 1 indicate strong discrimination.

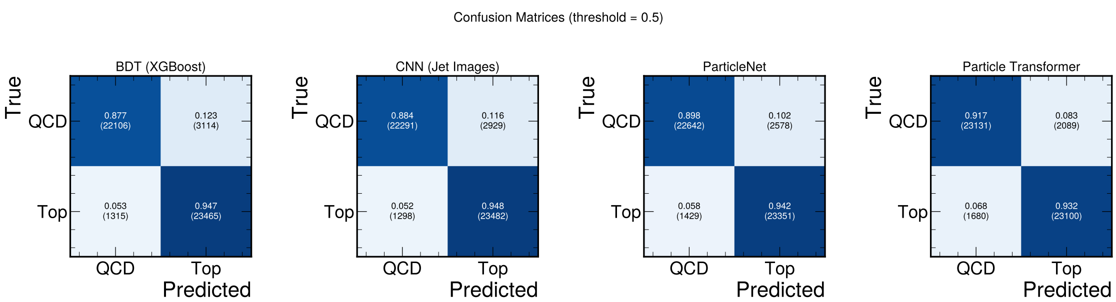

Figure 4.5: Confusion matrices (normalised) at a decision threshold of 0.5.

4.2 | Data-Efficiency Study

Figure 4.6 shows the test AUC and background rejection as a function of the training-set size N_{train}. Three findings stand out:

1. **The BDT is the most data-efficient model.** Its AUC rises by under one percent (from 0.961 to 0.968) across the whole 5k–50k range; engineered features short-circuit much of what the deep models must learn from data.

2. **The deep models show a clear crossover.** The Particle Transformer is the *weakest* model at 5k jets (AUC $= 0.949$) but the *strongest* at 50k (0.974), overtaking both the BDT and ParticleNet. Its slope is still positive at the largest size tested, consistent with the literature observation that it only saturates near 10^6 jets (Qu et al., 2022).

3. **The CNN is intermediate**, gaining substantially with data ($0.947 \rightarrow 0.968$) but plateauing earlier than the permutation-equivariant graph and attention models.

The practical message is that under a tight Monte Carlo budget the simple, interpretable BDT is competitive or better, whereas the attention model only realises its advantage once enough data is available.

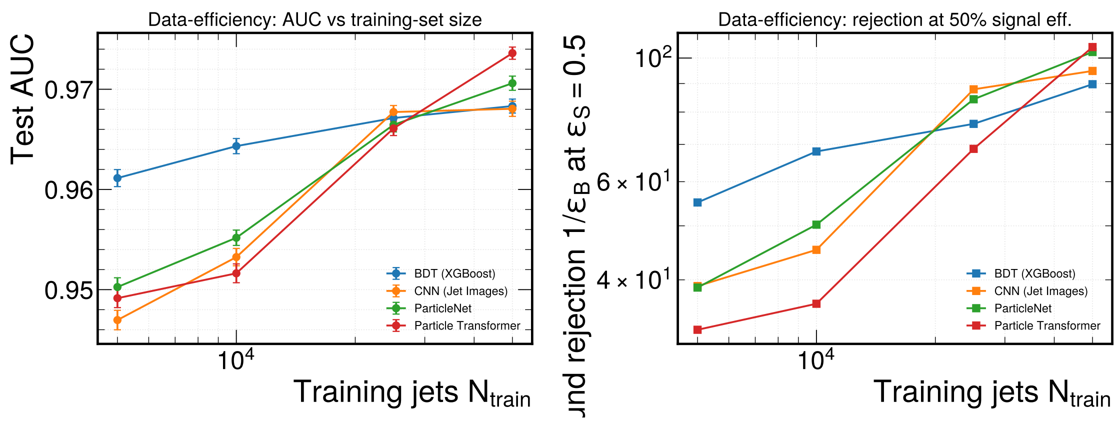

Figure 4.6: Data-efficiency curves. Left: test AUC versus training-set size with 1σ bootstrap error bars. Right: background rejection at $\varepsilon_S = 0.5$. The Particle Transformer rises from worst at 5k jets to best at 5×10^4, crossing the other models.

4.3 | Probability-Calibration Study

Figure 4.7 shows the reliability diagrams and score histograms, and Table 4.2 reports the ECE, the fitted temperature T and the post-rescaling ECE. The key results are:

1. **All models are mildly over-confident** (every fitted temperature satisfies $T > 1$), in line with Guo et al. (2017).
2. **Ranking and calibration are decoupled.** The jet-image CNN is the worst-calibrated model by an order of magnitude (ECE $= 0.078$, $T = 1.54$) despite ranking only third on AUC, whereas the top-ranked Particle Transformer is comparatively well-calibrated (ECE $= 0.012$). The BDT and ParticleNet are already well-calibrated (ECE ≈ 0.007).
3. **Temperature scaling helps, but not uniformly.** A single scalar brings the BDT, ParticleNet and Particle Transformer to ECE$\lesssim$ 0.008, but only halves the CNN's error (to 0.064): the CNN's miscalibration is structural and would need a richer recalibration such as isotonic regression.

Table 4.2: Test-set calibration at the 5×10^4-jet study endpoint. T is the temperature fitted on the validation split; ECE$_T$ is the Expected Calibration Error after rescaling.

Model	Test AUC	ECE	T	ECE$_T$
BDT (XGBoost)	0.9683	0.007	1.04	0.006
CNN (Jet Images)	0.9681	0.078	1.54	0.064
ParticleNet	0.9706	0.008	1.05	0.008
Particle Transformer	0.9736	0.012	1.12	0.007

4.4 | Model Interpretability

Two views of *why* the models work are useful. Figure 4.8 shows the BDT feature importances: the jet mass, τ_{32} and constituent multiplicity dominate, exactly the

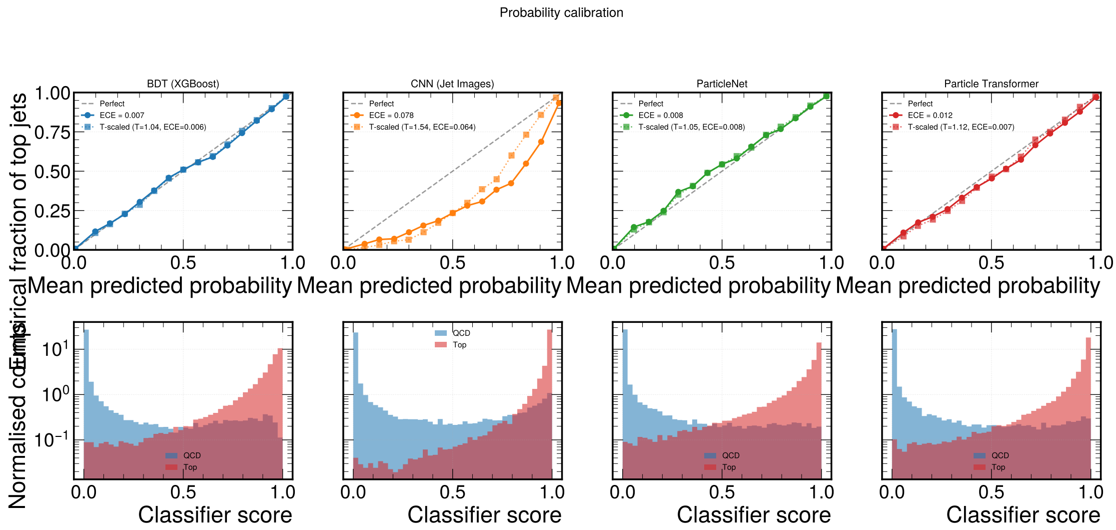

Figure 4.7: Top: reliability diagrams for the four models (solid: raw scores; dotted: after temperature scaling). Bottom: score histograms. The jet-image CNN deviates most strongly from the diagonal.

physically expected discriminators. Figure 4.9 shows the class-token attention of the Particle Transformer versus a constituent's angular distance ΔR from the jet axis. For QCD jets the attention concentrates at small ΔR (a single core), whereas for top jets it is spread to much larger ΔR, reflecting the broader, multi-pronged energy flow of the top decay. The model has learned to attend to the peripheral radiation that carries the top signature.

4.5 | Summary of Findings

1. The pipeline reproduces the established ordering Particle Transformer $>$ ParticleNet $>$ CNN $>$ BDT, with the Particle Transformer reaching AUC $= 0.978$.

2. Data efficiency separates the models sharply: the BDT is best in the small-data regime, while the deep models overtake it as data grows, producing a clear crossover.

3. Ranking power and calibration are decoupled: the CNN is the worst-calibrated

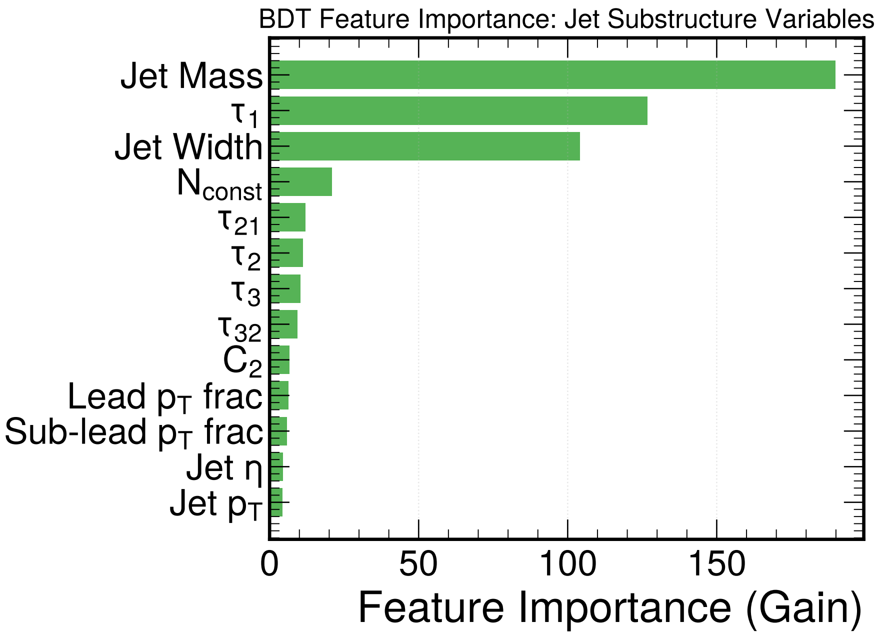

Figure 4.8: Gain-based feature importance of the BDT. The jet mass, τ_{32} and constituent multiplicity are the most discriminating variables.

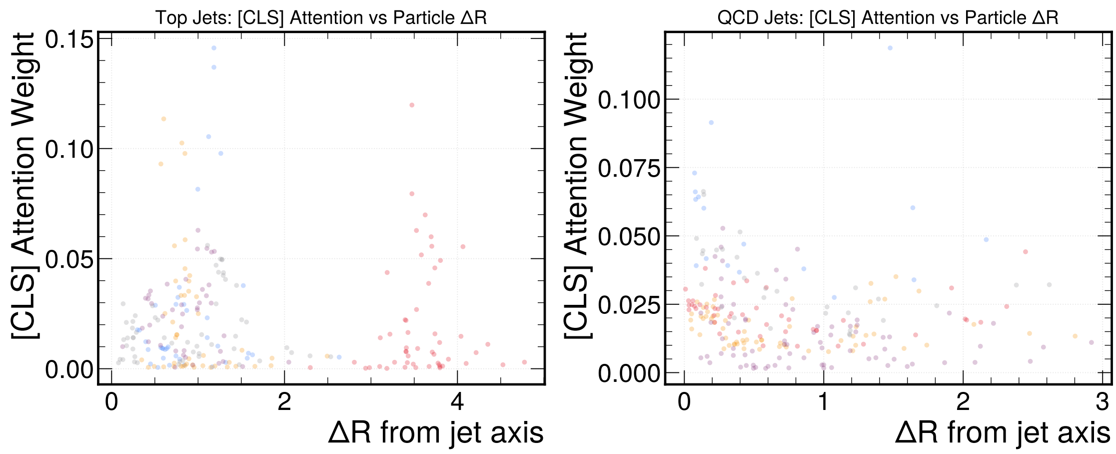

Figure 4.9: Class-token attention weight of the Particle Transformer versus the constituent angular distance ΔR from the jet axis, for top jets (left) and QCD jets (right). Top jets draw attention to larger ΔR, matching their multi-prong structure.

model despite a mid-pack ranking, and temperature scaling fixes every model except the CNN.

5

Conclusions and Future Work

5.1 | Conclusions

This thesis built a transparent, reproducible top-tagging pipeline and used it to compare four classifiers — a boosted decision tree on engineered features, a jet-image CNN, the ParticleNet graph network and the Particle Transformer — on the public 14 TeV Top-Quark Tagging Reference Dataset, with a focus on data efficiency and probability calibration rather than on a new record. Returning to the objectives of Chapter 1, the study reached the following conclusions.

1. **The established ordering is reproduced.** Under identical preprocessing and a modest budget, the Particle Transformer is best (AUC = 0.978, $1/\varepsilon_B \approx 136$ at $\varepsilon_S = 0.5$), followed by ParticleNet, the CNN and the BDT, matching the community benchmark.

2. **Data efficiency separates the models.** The BDT is the most data-efficient and is the strongest model at the smallest training size, but it is overtaken by the deep models as data grows. The Particle Transformer in particular rises from the weakest model at 5×10^3 jets to the strongest at 5×10^4, a clear crossover with direct practical implications for low-statistics analyses.

3. **Ranking and calibration are decoupled.** The highest-AUC model (the Particle Transformer) is comparatively well-calibrated, whereas the jet-image

CNN — only third on AUC — is the worst-calibrated by an order of magnitude. A single-parameter temperature rescaling restores good calibration for the BDT, ParticleNet and the transformer, but only partially corrects the CNN. A high AUC therefore does not by itself license the use of a model's score as a probability.

4. **World-class benchmarks are reproducible on free hardware.** The entire study was run on a free cloud GPU using open data and open tools, with automatic checkpointing and resumption, demonstrating that such work is accessible to students in resource-constrained settings.

5.2 | Limitations

The study used subsamples of up to 5×10^4 jets rather than the full 1.2×10^6, so absolute performances are slightly below the published saturation values; the trends, not the records, are the contribution. The reference dataset contains no pile-up and uses a single detector parametrisation, so detector robustness and high-luminosity behaviour are not addressed. Each architecture was trained with a fixed, modest configuration rather than an exhaustive hyper-parameter search, and four models, while representative, are not exhaustive.

5.3 | Future Work

Several extensions follow naturally from this work:

1. **Scale up.** Repeat the data-efficiency sweep up to the full 1.2×10^6 jets on a larger GPU to locate precisely where the transformer overtakes ParticleNet by a statistically significant margin.

2. **Richer recalibration.** Compare temperature scaling with non-parametric isotonic regression, particularly for the CNN whose miscalibration temperature scaling does not fully remove.

3. **Robustness.** Study calibration and discrimination under simulated pile-up and detector smearing, which is where calibration matters most for real analyses.

4. **More architectures.** Add a Deep Sets baseline and a Lorentz-equivariant network (Gong et al., 2022) to the comparison.

5. **Uncertainty quantification.** Use deep ensembles or Monte-Carlo dropout to attach predictive uncertainties to the tagger scores.

In summary, this project shows that asking *how much data* and *how trustworthy the probabilities are*, rather than only *how high the AUC*, yields a more complete and more useful picture of deep jet-tagging models — and that this picture can be obtained reproducibly on freely available hardware.

A

Reproducibility and Supplementary Material

A.1 | Code and Data Availability

The complete pipeline — data loading, feature engineering, the four models, training, evaluation and all plotting — is released as open source at

`https://github.com/ramc77/fyp-jet-tagging`

The dataset is the public Top-Quark Tagging Reference Dataset (Kasieczka et al., 2019a), available from Zenodo (`https://zenodo.org/records/2603256`).

A.2 | How to Reproduce the Results

On a machine with Python 3.11 the headline results and study are reproduced with:

```
python3 -m venv particle_venv
source particle_venv/bin/activate
pip install -r requirements.txt
python3 download_data.py --yes       # fetch the dataset
python3 run_full_pipeline.py         # train + evaluate models
python3 run_study.py                 # data-efficiency + calibration
```

A Google Colaboratory notebook (`notebooks/FYP_Colab.ipynb`) runs the same pipeline on a free T4 GPU and downloads the results. Every training stage checkpoints after each epoch and resumes automatically, so a run may be interrupted and continued.

A.3 | Training Configuration

Table A.1: Principal hyper-parameters of the four models.

Model	**Key settings**	**Parameters**
BDT (XGBoost)	500 rounds, depth 7, lr 0.1	∼160 trees
CNN (jet images)	3 ResNet blocks, 64–256 ch.	1.26 M
ParticleNet	3 EdgeConv blocks, $k = 16$	0.37 M
Particle Transformer	3 blocks, 4 heads, dim 128	0.43 M

All models minimise binary cross-entropy with the Adam/AdamW optimiser, gradient clipping, a plateau learning-rate scheduler and early stopping on the validation AUC, as detailed in Section 3.4.

A.4 | Training History

Figure A.1 shows the training and validation loss and AUC curves for the three neural models, confirming smooth convergence without significant over-fitting within the epoch budget.

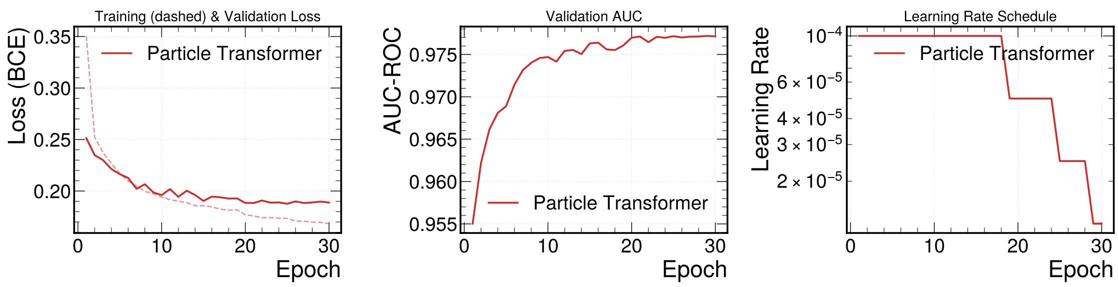

Figure A.1: Training/validation loss and validation AUC versus epoch for the CNN, ParticleNet and Particle Transformer.

A.5 | Transformer Attention Maps

Figure A.2 shows representative head-averaged self-attention maps for top jets across the transformer layers, complementing the class-token analysis in Section 4.4.

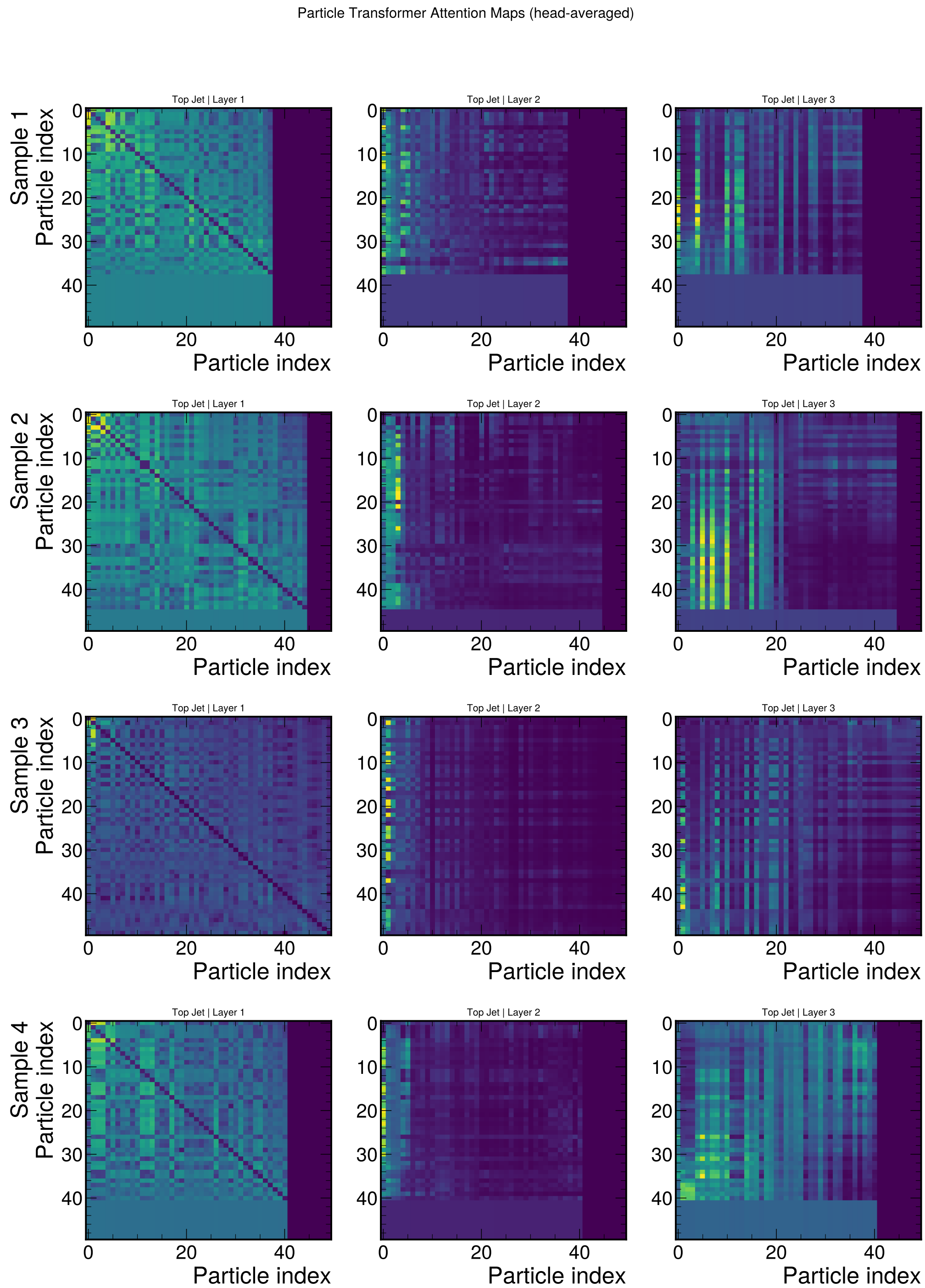

Figure A.2: Head-averaged self-attention maps of the Particle Transformer for sample top jets across the encoder layers. The padded (non-particle) positions appear as uniform blocks.

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